

Energy-efficient Analog Integrated Circuits

DCDC Converters – Exercise 1

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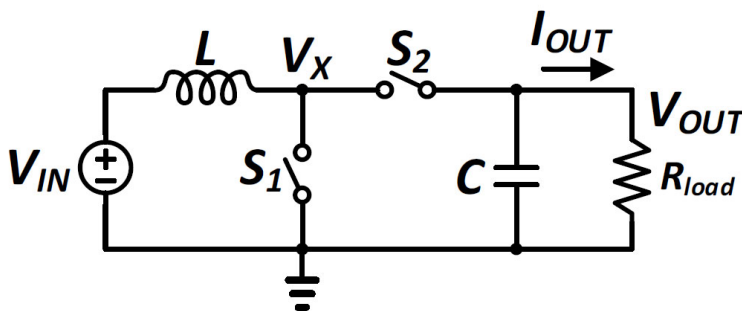
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Introduction

In this assignment you will analyze and design a Boost converter following the modeling you have learnt in the Tutorial. You can find the testbench files in `/home/eeaic_01fs26/DCDC_EEAIC_EX1`, you can copy the files to your local directory.

Part I – Modeling of the Boost Converter

Given a standard Boost converter topology and its specifications in the table:



V _{IN}	1.0V
V _{OUT}	2.0V
I _{OUT}	0.5A
L	470nH
RL (ESR_L)	40mOhm
C	1uF
f _{sw}	10MHz
R _{DS(on)}	80mOhm

1. (1 Pt) Draw circuit in phase 1 and phase 2
2. (3 Pt) Draw the inductor voltage and current waveforms in phase 1 and phase 2
3. (3 Pt) Use inductor voltage balance to determine $M=V_{OUT}/V_{IN}$
4. (6 Pt) Formulate and draw the capacitor C current waveform
5. (3 Pt) Use capacitor charge balance to determine the inductor average current
6. (3 Pt) Use inductor equation to calculate the inductor current slope
7. (3 Pt) Calculate inductor current ripple

8. (6 Pt) Calculate the V_{OUT} voltage ripple
9. (6 Pt) Include the inductor series resistance R_{ind} and reformulate M
10. (6 Pt) Plot M vs D in the 3 following cases:
 - a. $R_{load}=0.66$; $R_{ind}=0$;
 - b. $R_{load}=0.66$; $R_{ind}=R_{load}/100$;
 - c. $R_{load}=0.66$; $R_{ind}=R_{load}/20$;
 - i. For all three cases, please, use the following settings:
 1. $D=0.1:0.001:1$;
 2. $xlim([-0.05 \ 1.05])$
 3. $ylim([-0.05 \ 6.2])$
11. (6 Pt) Calculate the efficiency vs M including R_{ind} for $R_{load}=0.666$, $R_{ind}=R_{load}.*[0.01:0.001:0.05]$ and $D=0.5$
12. (3 Pt) Calculate and plot the inductor DC and AC conduction losses
13. (2 Pt) Calculate the S1 and S2 conduction losses P_{cond}
14. (3 Pt) Calculate the S1 and S2 switching losses P_{sw} accounting for their gate capacitance only, equal to $C_{gate}=500pF$ and $V_{gate,swing}=1.8V$
15. (3 Pt) Estimate the converter efficiency considering only P_{cond}
16. (3 Pt) Estimate the converter efficiency considering P_{cond} and P_{sw}

Part II – Simulation of the Boost Converter on Cadence

1. (10 Pt) Run the *TestBench_BOOST_idealSW* with $R_{ind}=1m\Omega$ and $R_{DSOn}=1m\Omega$, then include in your report the screenshots of the following variables plotted and show whether they match the previous calculation:
 - a. V_L , I_L , V_{OUT} , I_C , I_L ripple, V_{OUT} ripple, V_{OUT} average, I_L average
 - i. Make sure the y-axis is well readable and that the waveforms are clearly visible for all the signals
2. (5 Pt) Plot M vs $D=[0.2:0.1:0.9]$ for $V_{IN}=1.0V$ and $R_{load}=4$ in the following cases:
 - a. $R_L=1m$, $R_{DSOn}=1m$
 - b. $R_L=40m$, $R_{DSOn}=1m$
 - c. $R_L=40m$, $R_{DSOn}=80m$
3. (5 Pt) Plot Efficiency vs $D=[0.2:0.1:0.9]$ for $V_{IN}=1.0V$ and $R_{load}=4\Omega$ in the following cases
 - a. $R_L=1m$, $R_{DSOn}=1m$
 - b. $R_L=40m$, $R_{DSOn}=1m$
 - c. $R_L=40m$, $R_{DSOn}=80m$
4. (5 Pt) Use *nch_25* NMOS type transistors from the *tsmcN65* library for S1 and S2 and size them to achieve $R_{DSOn}=80m\Omega$ using minimum length transistors, what is the width you obtained?

For simplicity of the gate driving and high-side design, please, use the schematic below:

TestBench_BOOST_realSW/schematic

5. (5 Pt) Size S1 and S2 and size them to maximize the efficiency @ $V_{OUT}/V_{IN}=2.0V/1.0V$ considering the conduction losses only
6. (10 Pt) Repeat 5 considering S1 and S2 switching losses as well.